A Derivative Obstruction to Almost Periodicity in Variation

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Claimed status: full negative answer to the question in Example 3.3 of arXiv:2205.09031, subject to human review. For every $1 \le p < \infty$, the displayed function is Stepanov-p almost periodic, as recalled in the source, but it is *not* Stepanov-p almost periodic in variation.

1 Source question and definitions

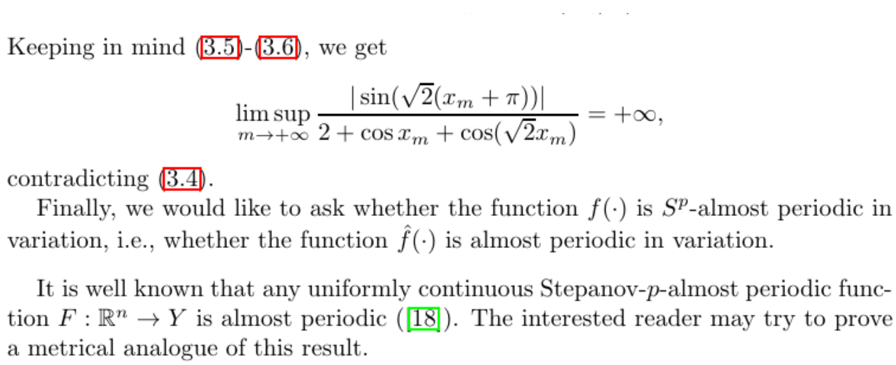
Set

$$f(t) = \sin\left(\frac{1}{\zeta(t)}\right), \qquad \zeta(t) = 2 + \cos t + \cos(\sqrt{2}t). \tag{1}$$

The source asks whether f is S^p-almost periodic in variation, meaning that its Bochner transform

$$\widehat{f}(t) \in L^p([0,1]), \qquad \widehat{f}(t)(s) = f(t+s),$$

is almost periodic in variation.



Keeping in mind (3.5)-(3.6), we get

$$\limsup_{m\to+\infty} \frac{|\sin(\sqrt{2}(x_m+\pi))|}{2+\cos x_m + \cos(\sqrt{2}x_m)} = +\infty,$$

contradicting (3.4).

Finally, we would like to ask whether the function $f(\cdot)$ is S^p-almost periodic in variation, i.e., whether the function $\hat{f}(\cdot)$ is almost periodic in variation.

It is well known that any uniformly continuous Stepanov-p-almost periodic function $F : \mathbb{R}^n \to Y$ is almost periodic ([18]). The interested reader may try to prove a metrical analogue of this result.

Figure 1: The question on PDF page 19 of arXiv:2205.09031v1.

For a continuous map $H : \mathbb{R} \to Y$ into a Banach space, write

$$\mathrm{Var}_1(H; q) = \sup_{q-1=t_0<\cdots<t_N=q+1} \sum_{j=0}^{N-1} \|H(t_{j+1}) - H(t_j)\|_Y.$$

The source's definition, specialized to ordinary almost periodicity, says that H is almost periodic in variation if, for every $\varepsilon > 0$, the set of τ satisfying

$$\sup_{q\in\mathbb{R}} \bigl(\|H(q+\tau) - H(q)\|_Y + \mathrm{Var}_1(H(\cdot+\tau) - H(\cdot); q)\bigr) < \varepsilon \tag{2}$$

is relatively dense.

2 The transfer lemma

Lemma 1 (Variation transfers to the derivative)**.** *Let $F \in C^1(\mathbb{R})$ and $1 \le p < \infty$. If $\widehat{F}(t) = F(t+\cdot)$ is almost periodic in variation as an $L^p([0,1])$-valued map, then F' is Stepanov-1 almost periodic. In particular,*

$$\sup_{q\in\mathbb{R}} \int_q^{q+1} |F'(u)|\, du < \infty. \tag{3}$$

Idea of the proof

The variation of a differentiable Banach-valued curve is the integral of the norm of its derivative. Differentiating a translation difference of the Bochner transform produces translation differences of F'. A short Tonelli argument then shows that the former variation dominates the usual Stepanov-1 distance between the latter functions.

Proof. Put $G = F'$ and, for a fixed τ, define

$$H_\tau(r) = \widehat{F}(r+\tau) - \widehat{F}(r).$$

On every compact r-interval this is a C^1 curve in $L^p([0,1])$, with

$$H_\tau'(r)(s) = G(r+s+\tau) - G(r+s).$$

Consequently,

$$\mathrm{Var}_1(H_\tau; q) = \int_{q-1}^{q+1} \|H_\tau'(r)\|_{L^p([0,1])}\, dr. \tag{4}$$

For every $q \in \mathbb{R}$, Tonelli's theorem and $\|h\|_1 \le \|h\|_p$ on the unit interval give

$$\begin{aligned}\int_q^{q+1} |G(u+\tau) - G(u)|\, du &\le \int_{q-1}^{q+1}\int_0^1 |G(r+s+\tau) - G(r+s)|\, ds\, dr \\ &\le \int_{q-1}^{q+1} \|H_\tau'(r)\|_p\, dr = \mathrm{Var}_1(H_\tau; q).\end{aligned} \tag{5}$$

Indeed, after the change $u = r+s$, the double integral has convolution weight exactly one for $u \in [q, q+1]$.

Thus every ε-translation in (2) is a Stepanov-1 ε-translation of G, and such translations are relatively dense. This proves Stepanov-1 almost periodicity.

For completeness, boundedness follows directly. Take $\varepsilon = 1$ and let L witness relative density. Given x, choose a 1-translation $\tau \in [x-L, x+L]$. Then

$$\int_x^{x+1} |G(u)|\, du \le 1 + \int_{x-\tau}^{x-\tau+1} |G(v)|\, dv,$$

and the final integral is bounded because $x - \tau \in [-L, L]$ and G is continuous. This proves (3). ☐

3 The specific function has unbounded derivative variation

The denominator in (1) is strictly positive. Equality to zero would force both cosines to equal -1, making $\sqrt{2}$ a ratio of odd integers. Hence $f \in C^\infty(\mathbb{R})$ and

$$g(t) := f'(t) = \frac{\sin t + \sqrt{2}\sin(\sqrt{2}t)}{\zeta(t)^2} \cos\left(\frac{1}{\zeta(t)}\right). \tag{6}$$

Lemma 2. *There is a fixed $b \in (0,1)$ and a sequence of intervals $[m_n, m_n + b]$ such that*

$$\int_{m_n}^{m_n+b} |g(t)|\, dt \longrightarrow \infty.$$

In particular, g is not Stepanov-1 bounded.

Proof intuition

The irrational orbit $(t, \sqrt{2}t)$ comes arbitrarily close to (π, π) modulo 2π. Near each such visit, ζ has a positive, strictly convex well whose minimum tends to zero. On one side of the well, the change of variables $u = 1/\zeta(t)$ turns the total variation of $\sin(1/\zeta)$ into a long integral of $|\cos u|$.

Proof. The sequence $(2k+1)\sqrt{2}$ is dense modulo 2. We may therefore choose integers k_n such that, for $x_n = (2k_n + 1)\pi$,

$$e^{i\sqrt{2}x_n} \longrightarrow -1.$$

It follows that

$$\zeta(x_n) \to 0, \qquad \zeta'(x_n) \to 0, \qquad \zeta''(x_n) \to 3.$$

More precisely, uniformly for $|h| \le a$,

$$\zeta''(x_n + h) \longrightarrow \cos h + 2\cos(\sqrt{2}h).$$

Choose a fixed, sufficiently small $a \in (0, 1/2)$. For all large n,

$$\zeta''(t) \ge 1 \quad (|t - x_n| \le a). \tag{7}$$

Since $\zeta'(x_n) \to 0$, strict convexity gives a unique $m_n = x_n + o(1)$ in this interval with $\zeta'(m_n) = 0$. It is the local minimum, and

$$0 < \delta_n := \zeta(m_n) \le \zeta(x_n) \longrightarrow 0.$$

Fix $b = a/2$. For large n, (7) holds throughout $[m_n, m_n + b]$; hence ζ is increasing there and

$$\zeta(m_n + b) \ge \delta_n + \frac{b^2}{2} \ge \frac{b^2}{2}.$$

Using (6) and then the monotone substitutions $v = \zeta(t)$ and $u = 1/v$, we obtain

$$\begin{aligned} \int_{m_n}^{m_n+b} |g(t)|\, dt &= \int_{\delta_n}^{\zeta(m_n+b)} \frac{|\cos(1/v)|}{v^2}\, dv \\ &= \int_{1/\zeta(m_n+b)}^{1/\delta_n} |\cos u|\, du \longrightarrow \infty. \end{aligned}$$

The lower endpoint is at most $2/b^2$, whereas the upper endpoint tends to infinity, and $|\cos u|$ has positive mean $2/\pi$. Since $b < 1$, these intervals lie inside unit intervals, proving the last assertion. □

4 Answer

Theorem 3 (Full negative resolution)**.** *For every $1 \le p < \infty$, the function in (1) is not S^p-almost periodic in variation.*

Proof. If it were, $\widehat{f}$ would be almost periodic in variation as an $L^p([0,1])$-valued map. Lemma 1 would make $f' = g$ Stepanov-1 bounded, contradicting Lemma 2. □

Remark 4 (Scope)**.** *The argument does not dispute the source's statement that f is ordinary Stepanov-p almost periodic. It separates that property from the strictly stronger variation version. Lemma 1 is also a general obstruction: any C^1 function with unbounded local L^1 derivative variation gives the same negative conclusion for every finite p.*

A bounded novelty search of the run registry, exact question wording, displayed formula, and the 2025 journal publication found no later explicit answer. Novelty is provisional; the proof above is self-contained.

References

[1] B. Chaouchi, M. Kostić, and D. Velinov, *Metrical approximations of functions*, arXiv:2205.09031v1 (2022); Funkcialaj Ekvacioj **68** (2025), 1–44, doi:10.1619/fesi.68.1.